Small-scale, big impact: Evaluating UMP membranes at the Ambr 250ht scale

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Accelerating upstream development with Ambr® 250ht

The Ambr 250ht system's advanced automation capabilities and small footprint allow us to…

Clone selection	Process Optimization	Process characterization
Screen more clones	Search a larger optimization space	Include more parameters per characterization block

Want to understand what filter usage and sieving look like at larger scales: Can we use the Ambr 250 to predict scale-up performance?

Slide Credit to Kevin McFarland

Prior experiments identified that filter dimensions and chemistry strongly impact filter fouling

Presented at 2024 Planova Conference by Andrea Squeri

Confirmed that **increasing diameter** and **reducing filter length** improve sieving and lower hydraulic membrane resistance

Hagen-Poiseuille:

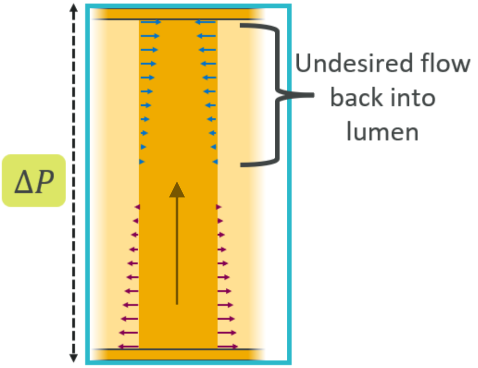
$$\Delta P = \frac{8\mu LQ}{\pi R^4}$$

Shear rate:

$$\gamma = \frac{4Q}{\pi R^3}$$

At constant shear rate:

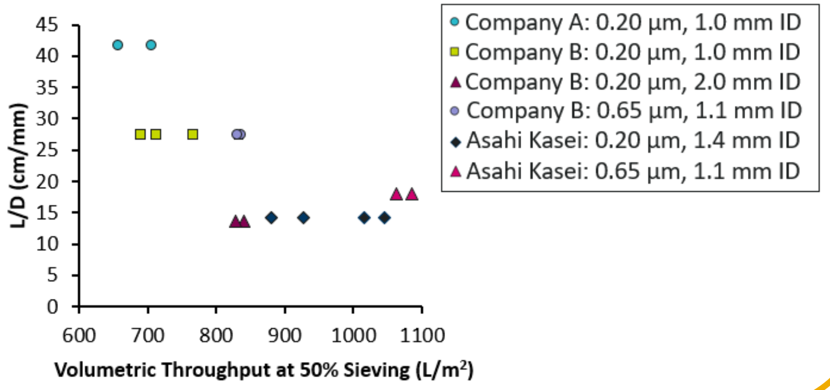
$$\Delta P \ \alpha \ \frac{L}{R}$$



Pore size important for membrane resistance but not sieving

Filters from Asahi Kasei showed **improved performance** over Company A and B due to a combination of filter geometry/ **chemistry** at fluxes between 1-5 LMH

Manufacturer	Lumen ID	Fiber Length	Membrane Chemistry
Asahi Kasei	1.1-1.4	19.8	PVDF
Company A and B	various	various	PES & mPES



More details available in the publication:

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Effect of inner diameter, filter length, and pore size on hollow fiber filter fouling during perfusion cell culture

Dominique WuDunn, Andrea Squeri, Jimmy Vu, Ashna Dhingra, Jon Coffman, Ken Lee

First published: 11 February 2024 | https://doi.org/10.1002/btpr.3440 | Citations: 2

Figures courtesy of Andrea Squeri

UMP filters necessary at the Ambr 250ht perfusion scale to accurately predict performance at larger scales

Partnered with Asahi Kasei to test the application of UMP filters in the Ambr 250

Ambr 250 single-use perfusion TFF vessel

Default filter:

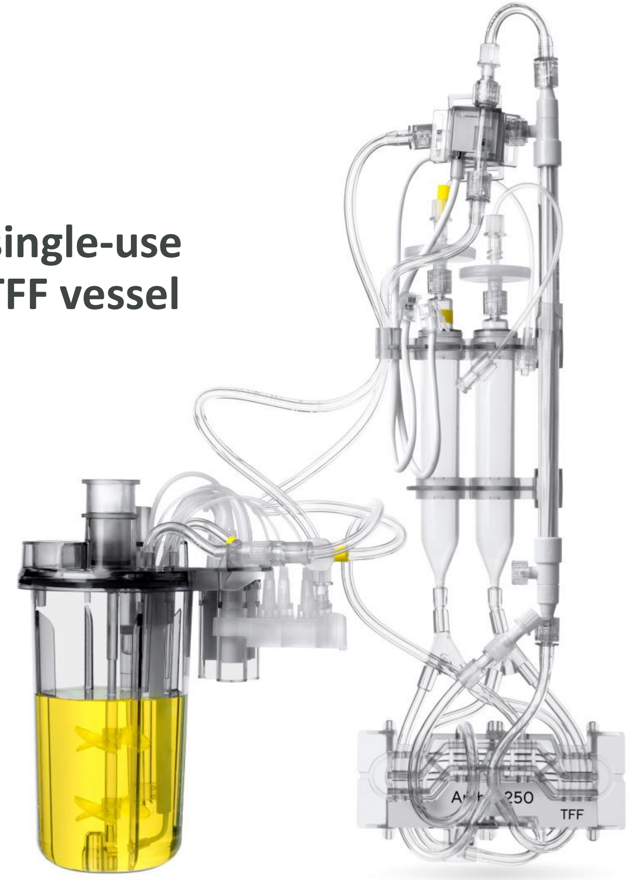
Area: 88cm^2

Fiber height: 20 cm

N Fibers: 14

Packed: Dry

Condition: Sterile



XUMP-0005W

Area: **50cm^2**

Fiber height: 22.3 cm

N Fibers: **5**

Packed: Water

Condition: Sterile

XUMP-0007W

Area: **70cm^2**

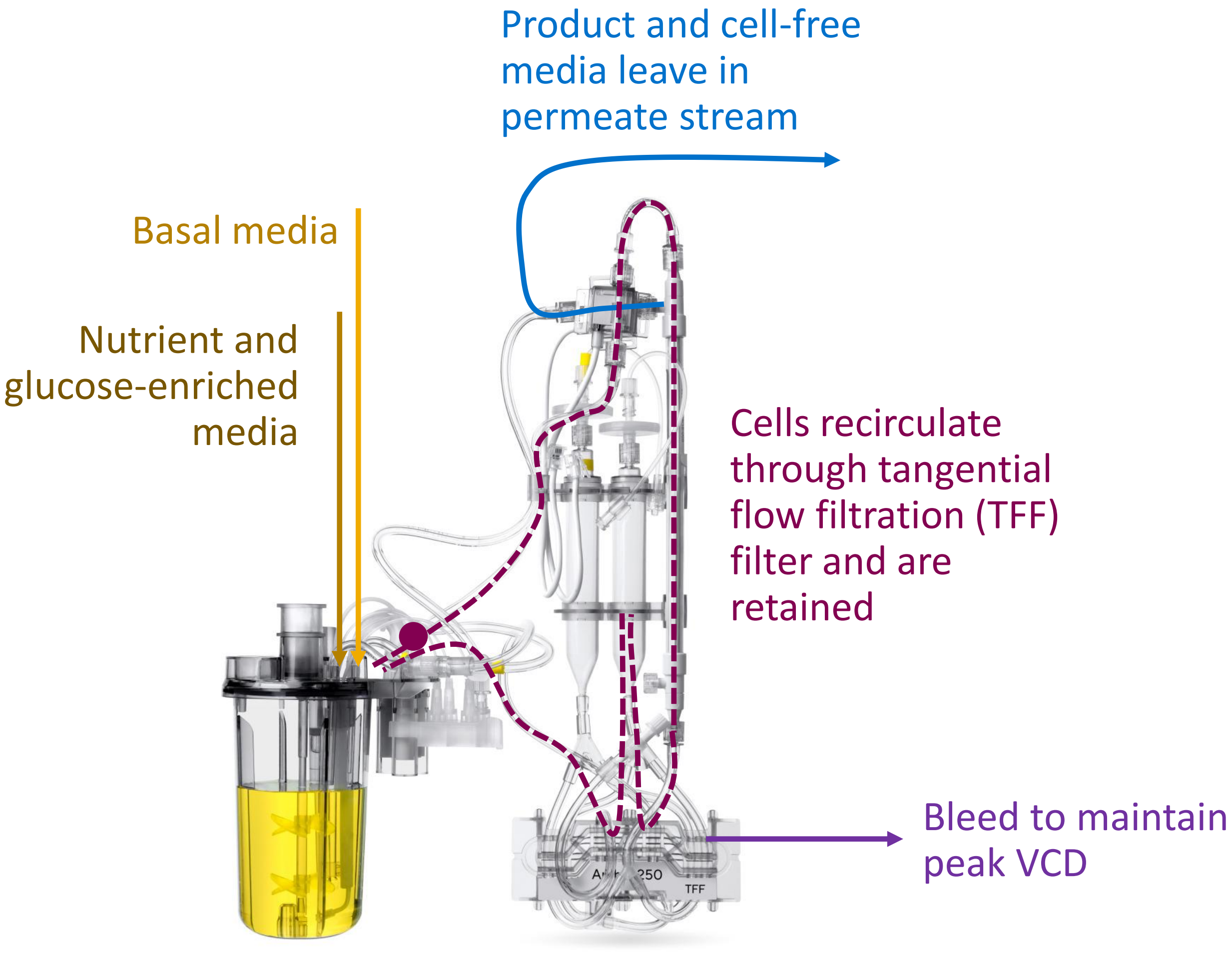
Fiber height: 22.3 cm

N Fibers: **7**

Packed: Water

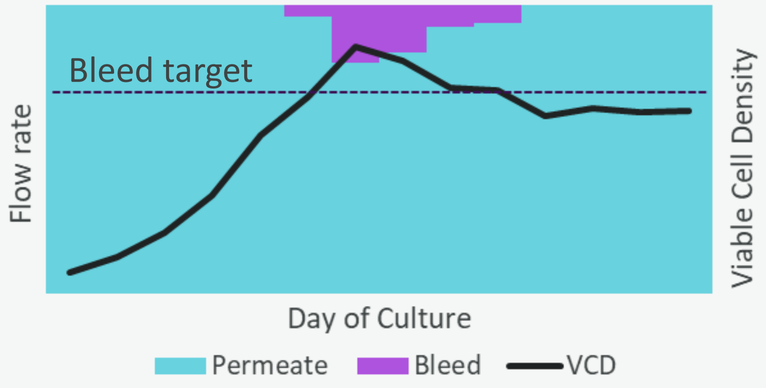
Condition: Sterile

Upstream CHO cell culture perfusion platform grows cells to high densities



Feeding ratio changed to control glucose concentration in the vessel and meet cell culture demand

Bleeding occurs after cell density or cell volume bleed target is reached; **perfusion set point** (1-1.5 VVD) maintained by reducing permeate flow rate

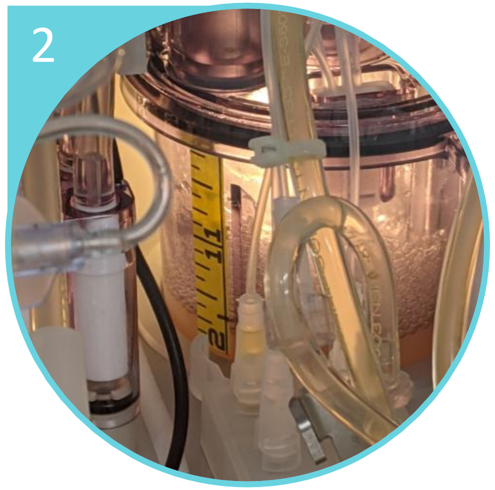


Goals of testing UMP filters

Successfully attach filters

Evaluate success based on:

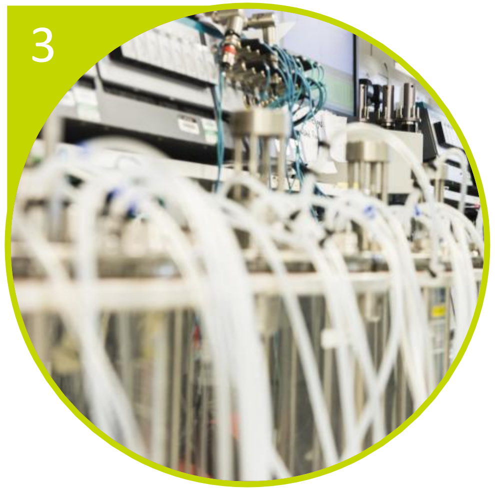
- Ease of attachment & Ergonomics
- Physical fit of filters
- Contamination risk



Observe performance

Identify:

- Impact on cell culture
- Rate and signs of filter failure



Determine Scale-Up Relevance

Compare performance and capabilities to 3-L bioreactors

Filters attached successfully mid-run with minor adjustments

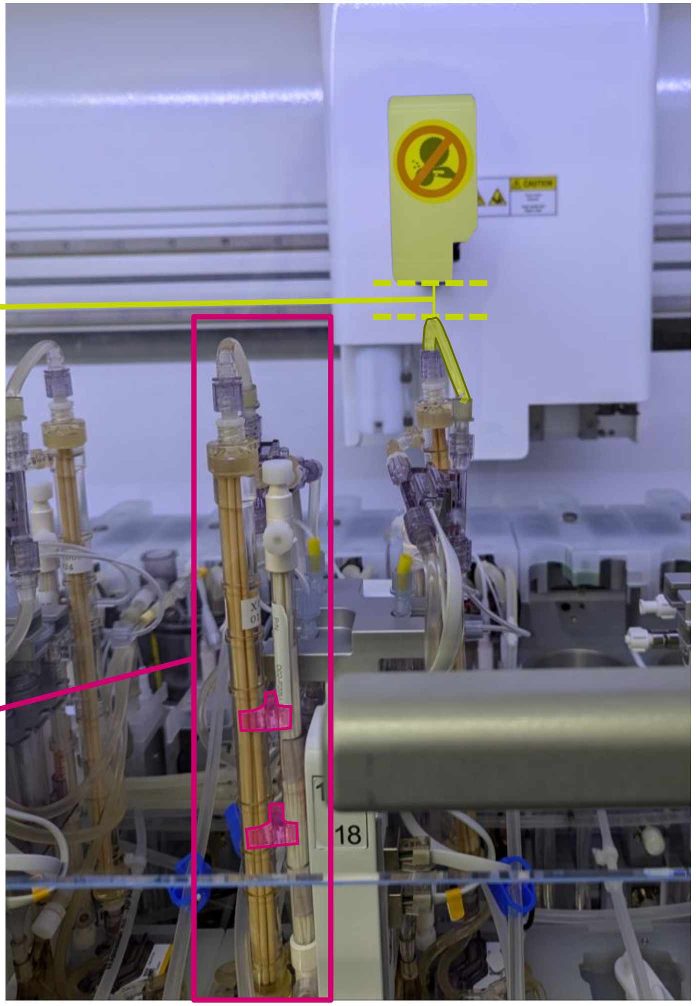
Filters attached after transition to production

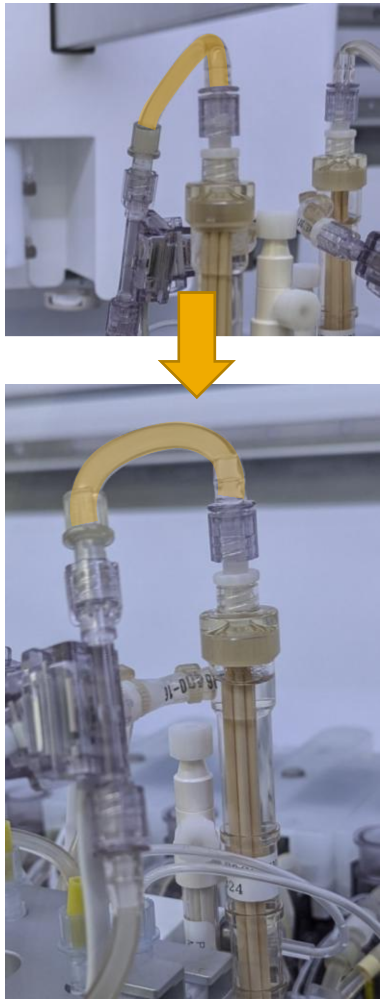
2 cm of clearance between the liquid handler arm and the top of the filter assembly

UMP **fibers are 2.3 cm longer** than original filter

UMP filter **housing is ~3-4 cm taller** than original filter housing and slightly wider

Clips onto existing filter housing





Some **careful adjustment** of the permeate and filter inlet tubing improves vertical slack and rounds the path to the filter

Experiment 1 Conditions
Run with Cell Line A

Cell line A produces a bispecific antibody and is typically a heavy fouler

1.0-1.4 VVD perfusion

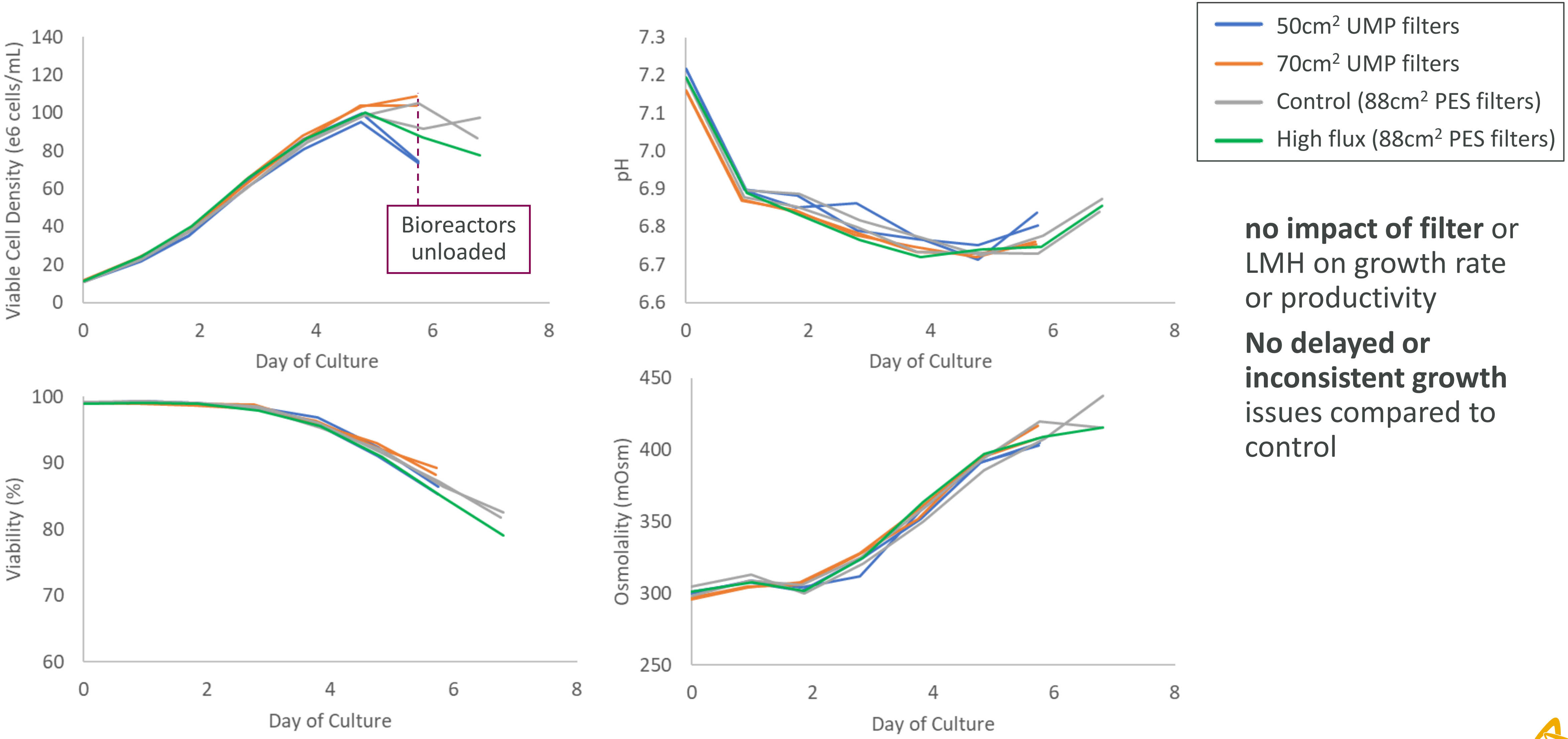
Working volume of 240 mL

Condition	Replicates	Filter pore diameter (µm)	Filter chemistry	Filter area (cm^2)	Packing Solution	Flux (LMH)	Fiber shear rate (1/s)
UMP Filter 50cm^2	2	0.2	PVDF	50	Water (sterile)	2.4	1480
UMP Filter 70cm^2	2	0.2	PVDF	70	Water (sterile)	1.7	1060
Control	2	0.2	PES	88	Dry	1.36	1454
Recycle loop/ high flux	2*	0.2	PES	88	Dry	3.5	1454

*1 replicate removed from dataset due to issue with sparger

No impact on cell culture performance observed

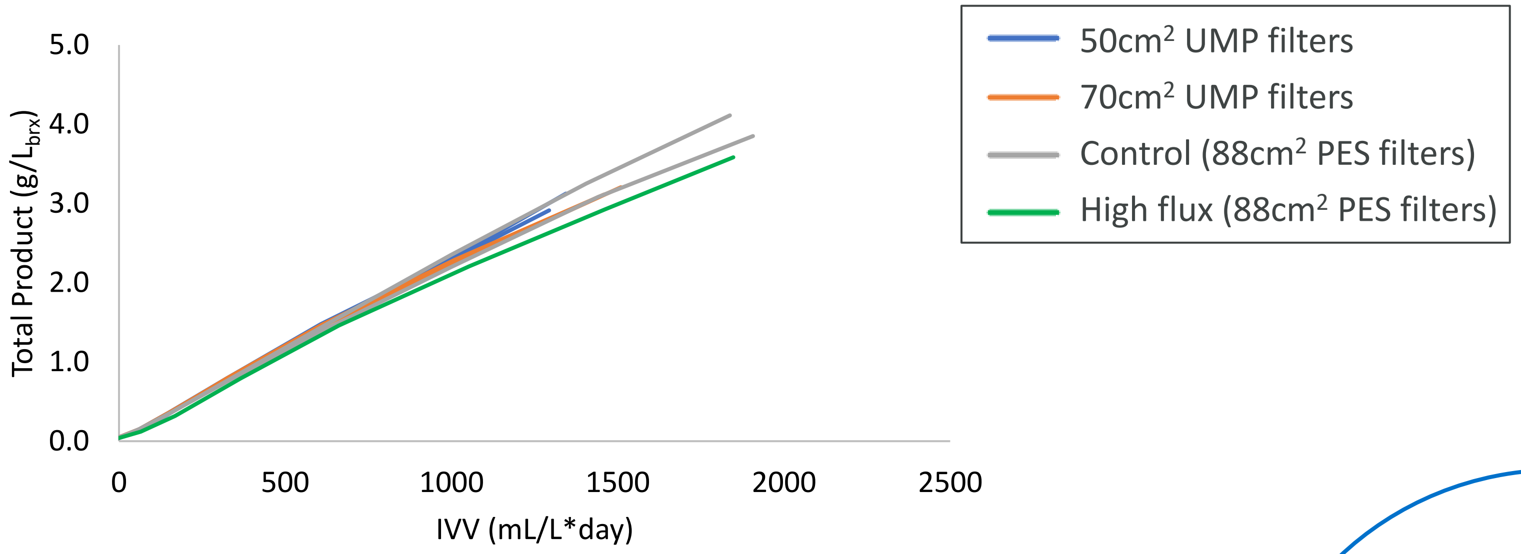
All UMP filter bioreactors unloaded after day 6 due to flux decay



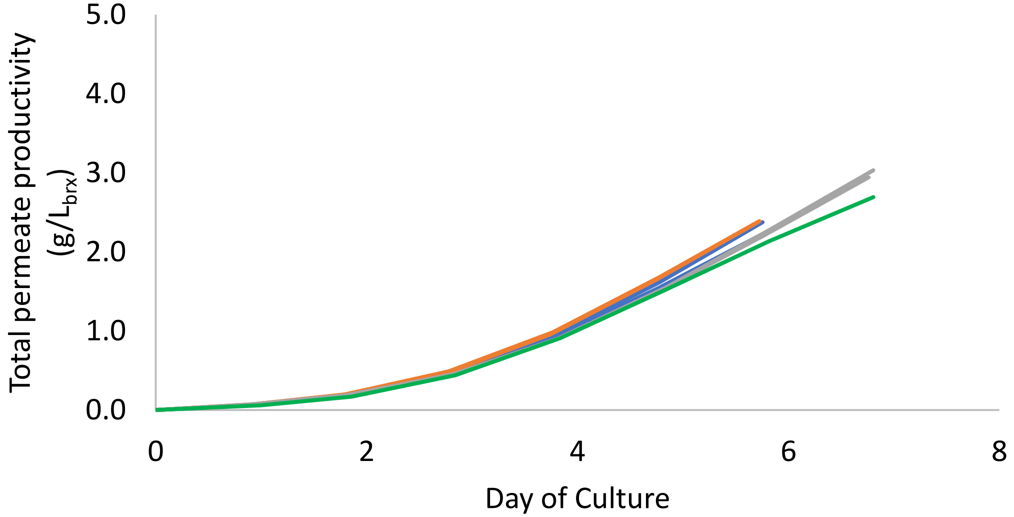
no impact of filter or LMH on growth rate or productivity

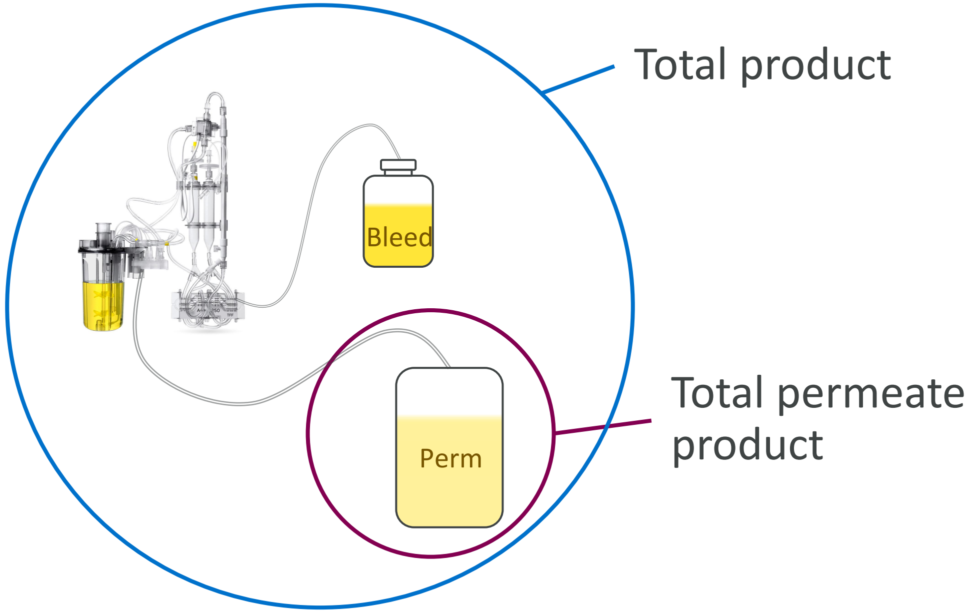
No delayed or inconsistent growth issues compared to control

No impact on cell volume-specific productivity



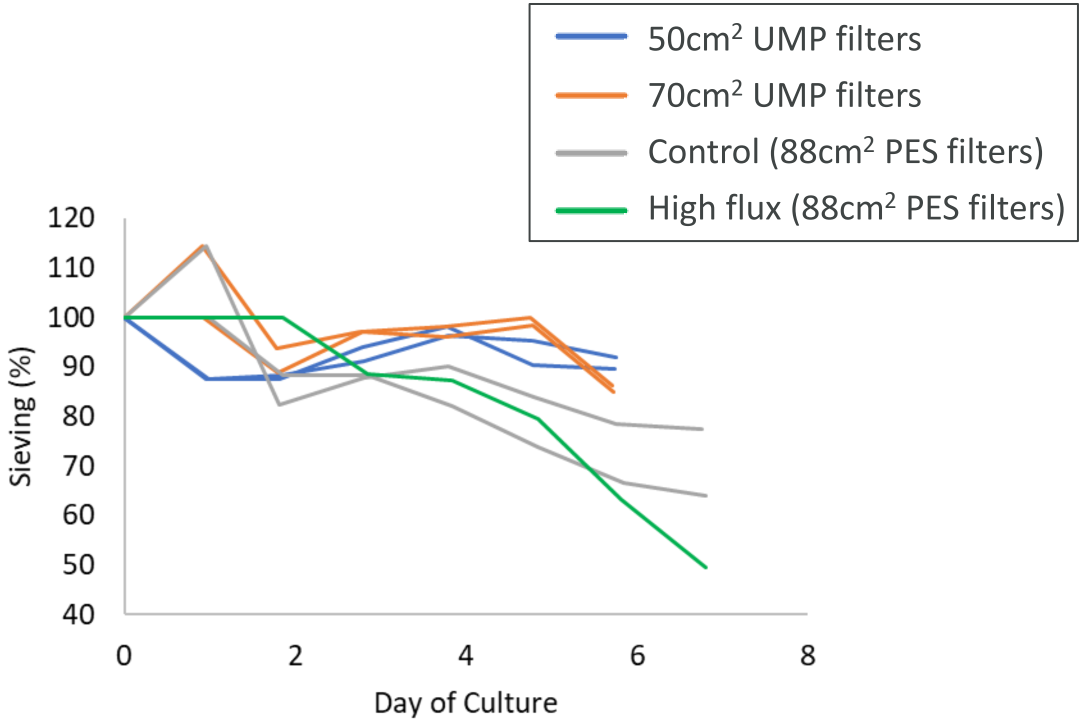
no impact on titer production per cell volume

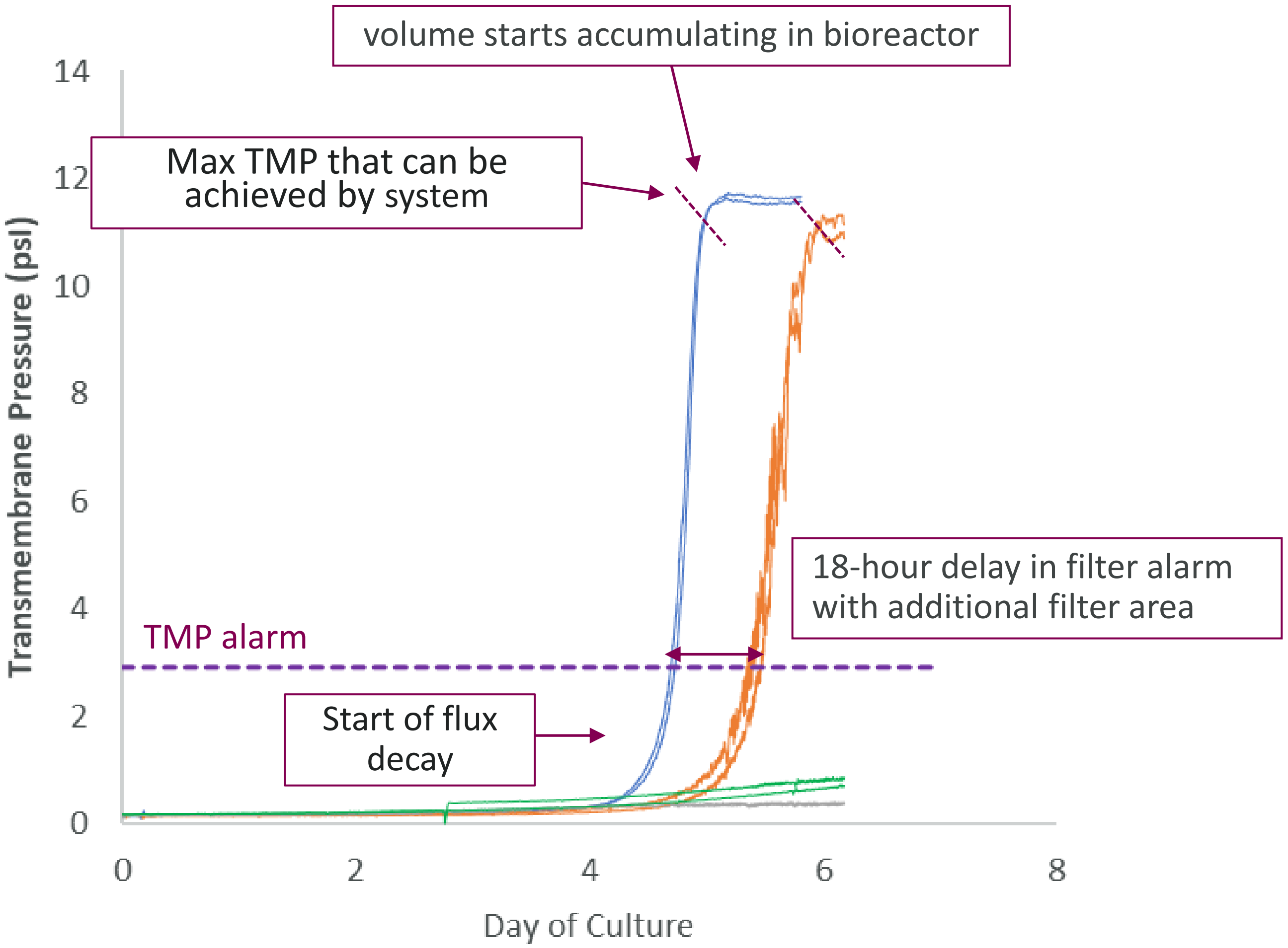




UMP filters showed 100% sieving until flux decay

Flux decay caused experiment to end early





- High sieving for UMP filters
- Filter blockage starts on evening of day 5 evening for 50cm^2 and evening of day 6 for 70cm^2
- No replacement filters available: all UMP filter bioreactors unloaded evening of day 6

High Flux testing possible in Ambr 250 setup

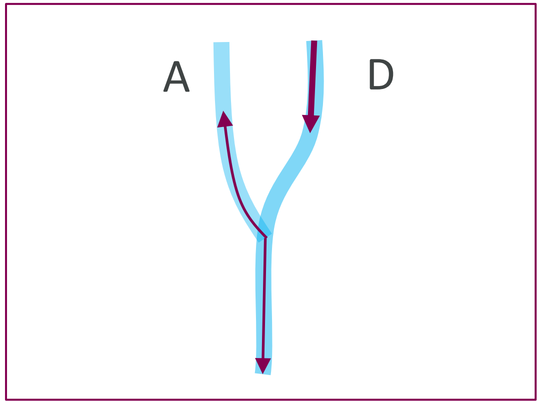
How can we mimic different ratios of filter size to bioreactor size?

Set up:

- Custom Y-tubing connects pump D (permeate out) and pump A (permeate return)
- In the protocol, the vessel volume exchange rate (VVD) is dictated by desired high flux
- Pump D pumps out more permeate than will reach the permeate collection container; pump A returns the excess permeate to the bioreactor

Considerations:

- When the permeate pump stops pumping during cell retention device calibrations, pump A pulls liquid back up the line
- Higher contamination risk due to more complicated set up and return of permeate liquid
- Custom tubing required



How can we predict filter performance across scales?

In-house hybrid mechanistic model combines experiment data, filter geometry, and fluid flow equations to predict filter fouling and flux decay across scales

Model Details

Cell growth kinetics determine particle size distribution from culture data

Flow models use serial filtration approach fluid mechanics to determine flux and pressure profiles

Filter fouling equations and **convective and diffusive hindrance effects** determine membrane permeability and available area

Model Training

Model **sees full Ambr 250 experimental data:** filter sieving & pressure, cell culture, flow rates

Model **sees Ambr 250 filter parameters**: number of filters & filter geometry, recirculation rate, flux

Model **fits filter fouling mechanism coefficients*** from data

* unique to cell line-membrane chemistry combination

Model Prediction

Model **sees partial experimental data at scale*:** cell culture, flow rates

Model **sees at-scale filter parameters**: number of filters, filter geometry, recirculation rate, flux

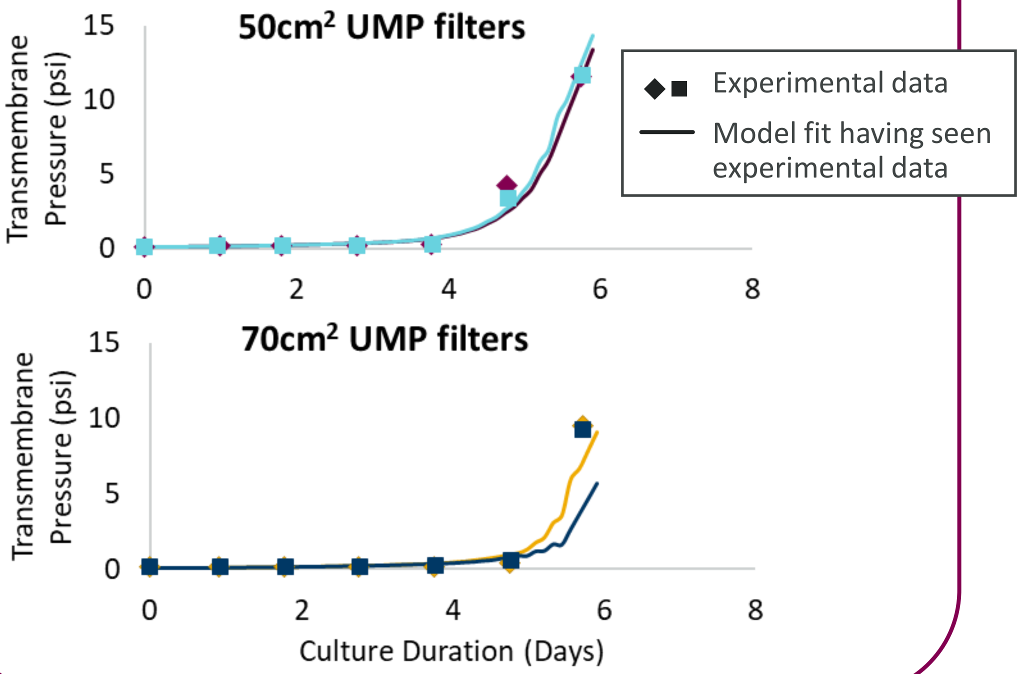
Model uses coefficients from Ambr 250 data to **predict filter fouling at scale**

Filter flux decay at 3-L scale was not consistently predicted for Cell Line A

1. Experimental data is ingested by the model to generate filter fouling mechanism coefficients

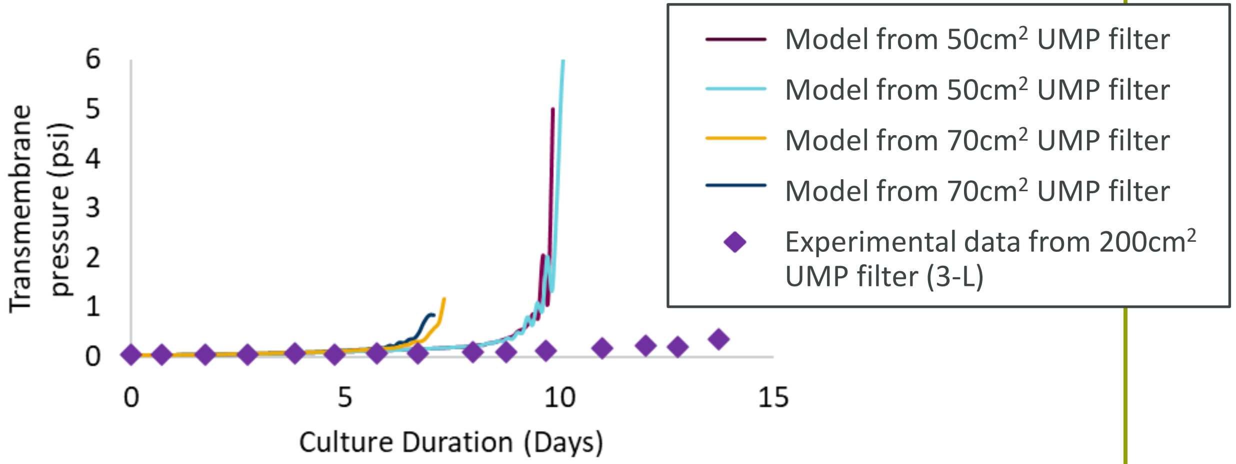
One set of coefficients is generated per bioreactor

These model fits have "seen" the experimental data



2. Can we predict across scales using models trained on Ambr 250 data?

When predicting filter fouling for a 3-L experiment, all Ambr 250-trained models predict earlier filter failure than reality



Early decision to run at Ambr 250 platform recirculation rate led to **elevated shear rates** that could affected axial flux distribution and potentially shift the primary fouling mechanism

- Typical 3L fiber shear rate = 303 1/s
- 50cm^2 fiber shear rate = 1480 1/s
- 70cm^2 fiber shear rate = 1060 1/s

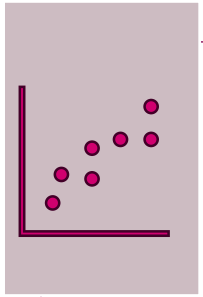
Some concerns that **filters became un-wetted** prior to use

Experiment 1 Conclusions and Experiment 2 Goals

UMP filters can be attached to the Ambr 250 TFF vessel with minimal risk of contamination

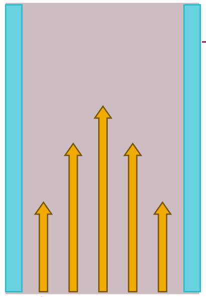
→ Swap filters mid-experiment

keep filters as wetted as possible prior to use



No impact on culture performance observed, and sieving profiles were consistent with experiments performed at other scales

→ Confirm on secondary cell line



Filter flux decay was not representative, potentially due to shear rate settings.

→ Test shear rate more representative of large-scale filters.

Experiment 2 focuses on testing a second cell line with more representative shear rates

Run with Cell Line B

Cell line B produces a monoclonal antibody and is typically a moderate fouler

1.0-1.4 VVD perfusion

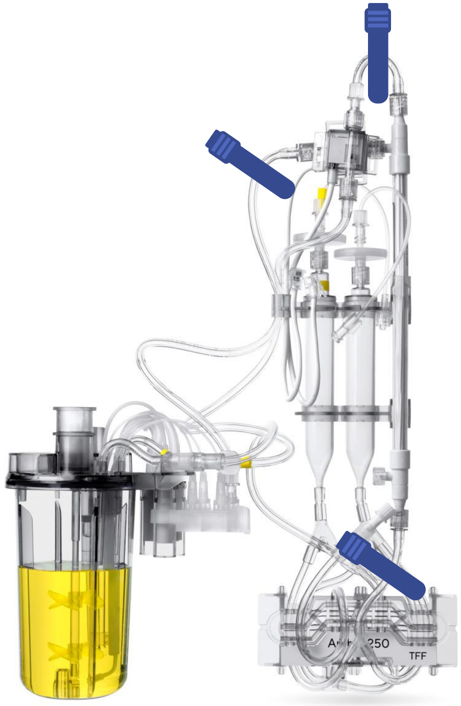
Working volume of 240 mL

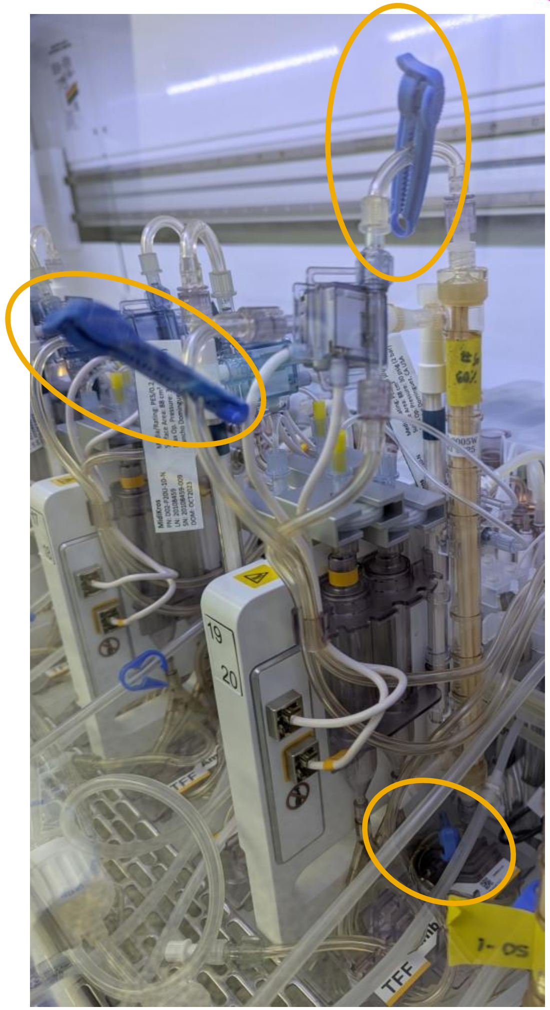
Condition	Replicates	Filter pore diameter (µm)	Filter chemistry	Filter area (cm^2)	Packing Solution	LMH	Fiber shear rate (1/s)
UMP Filter 50cm^2	2	0.2	PVDF	50	Water (sterile)	2.4	616
UMP Filter 70cm^2	2	0.2	PVDF	70	Water (sterile)	1.7	442
Control	2	0.2	PES	88	Dry	1.36	606

Filters successfully attached pre-perfusion

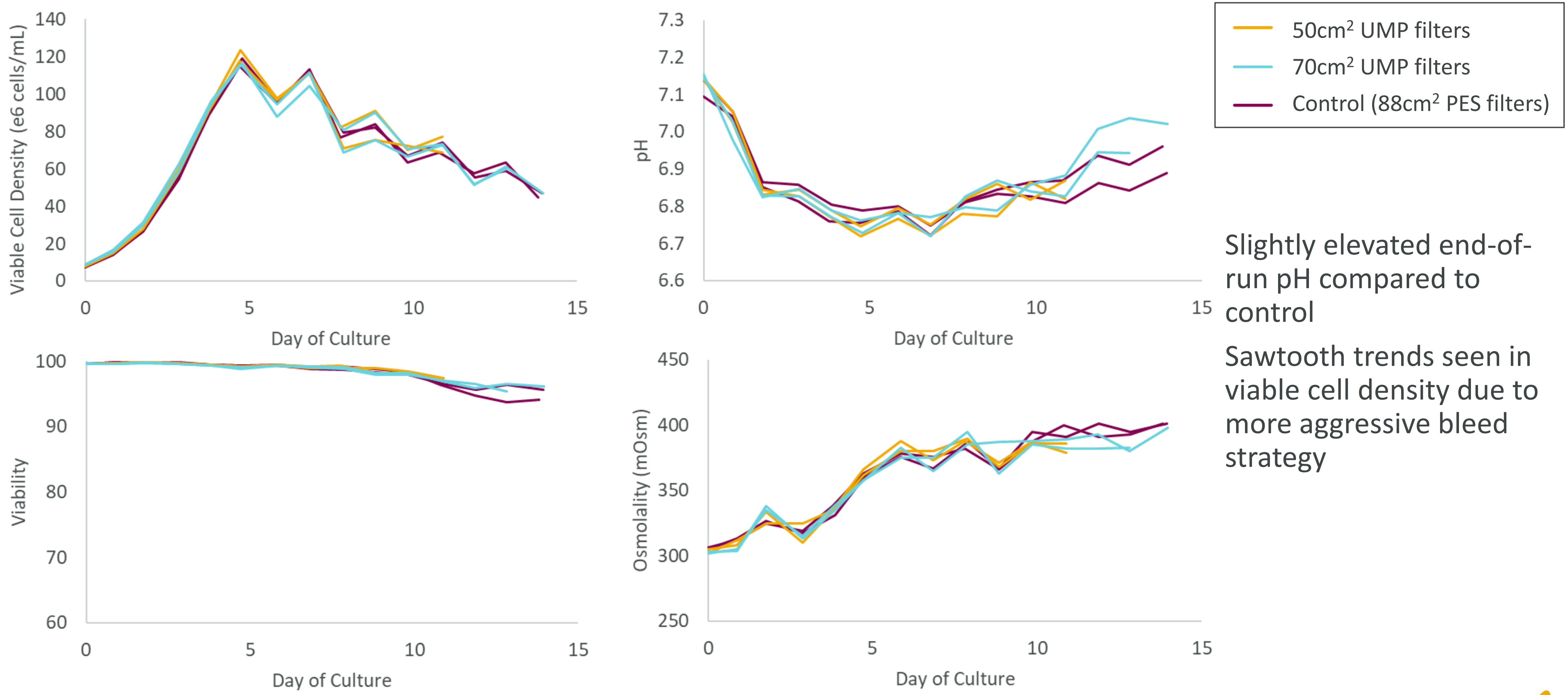
Filters kept wetted until recirculation started

- Filters attached in-place after bioreactor install due to timing of validation testing
- Clamps placed on all tubing connected to the filters to ensure filters remained wetted
- Clamps kept in place until recirculation started





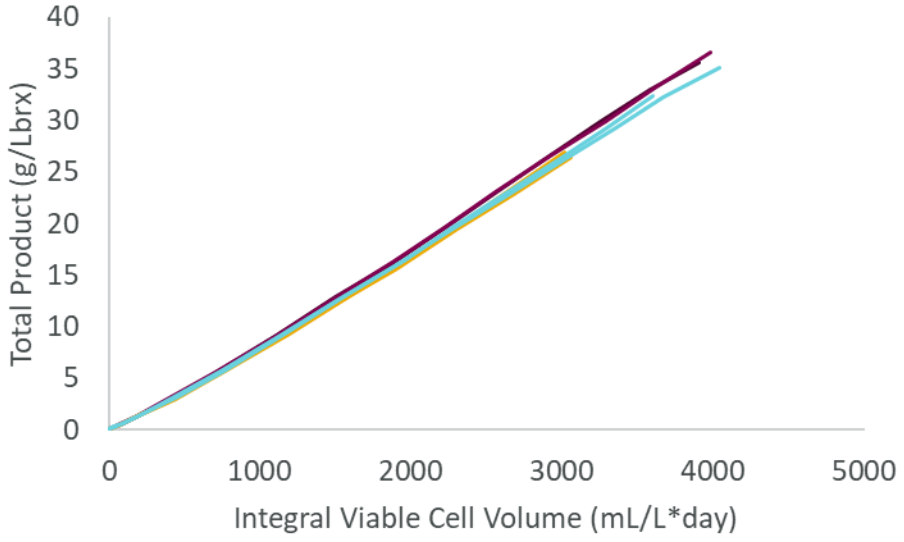
No impact on cell culture performance for Cell Line B

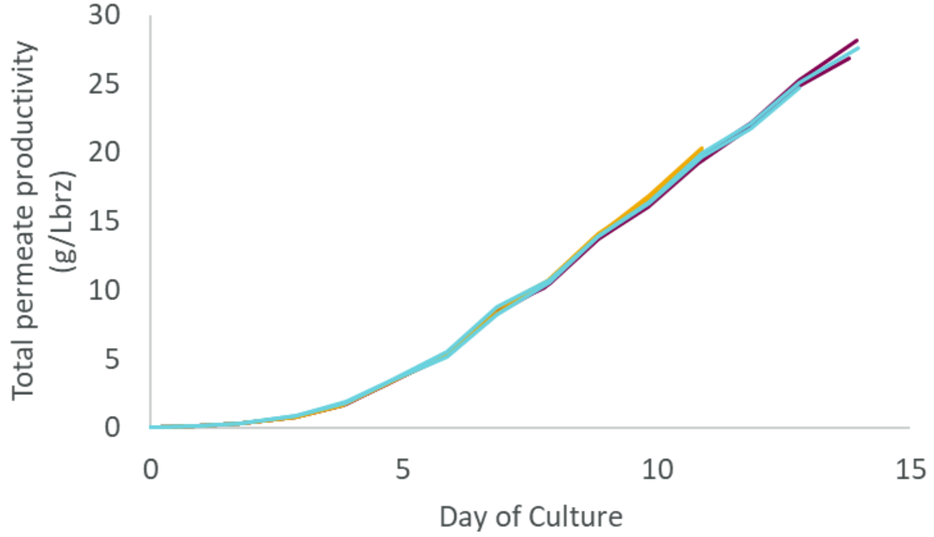


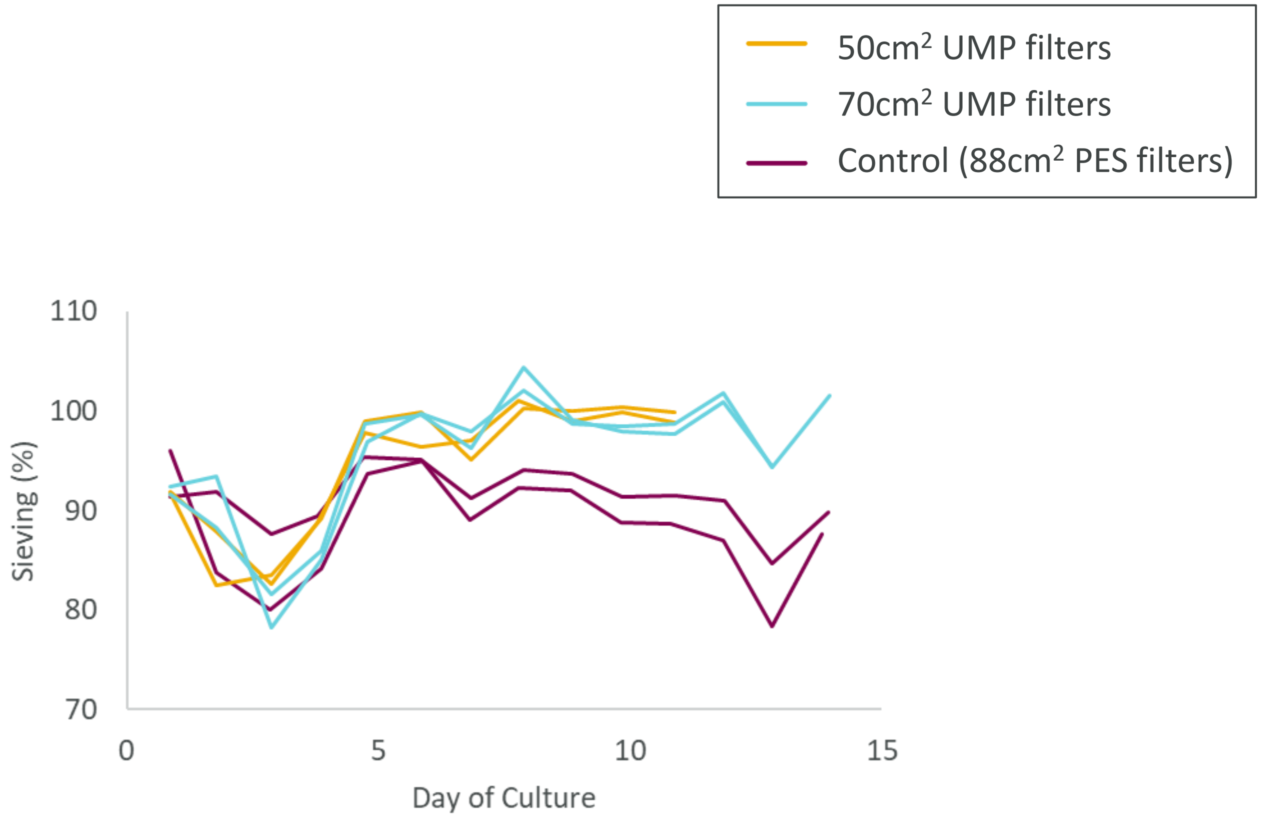
Slightly elevated end-of-run pH compared to control

Sawtooth trends seen in viable cell density due to more aggressive bleed strategy

UMP filters show improved sieving and no impact on cell productivity

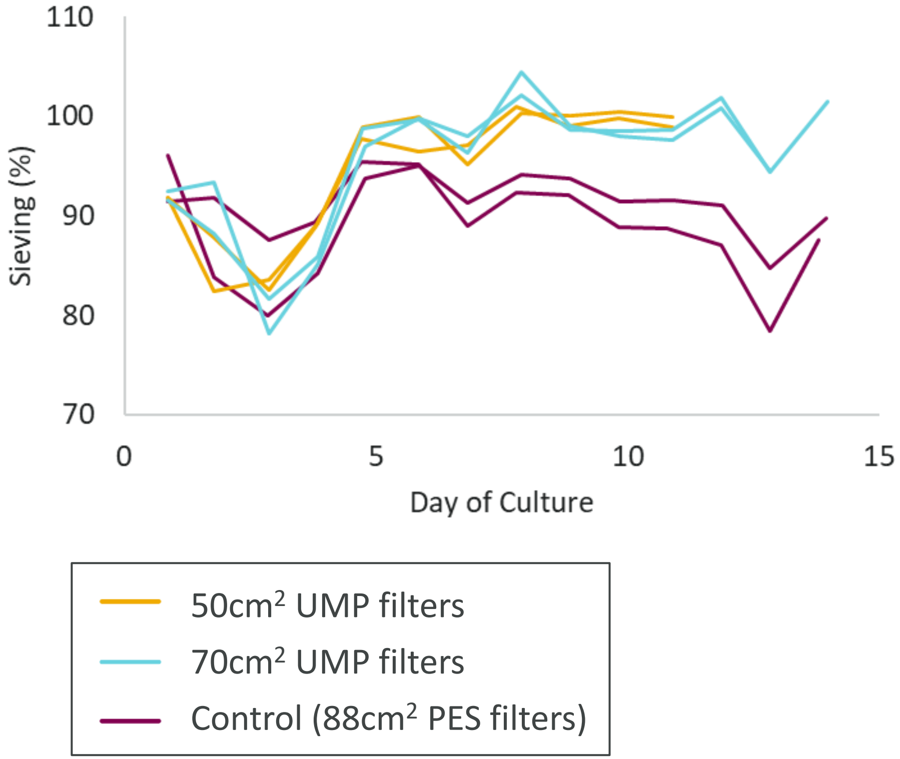


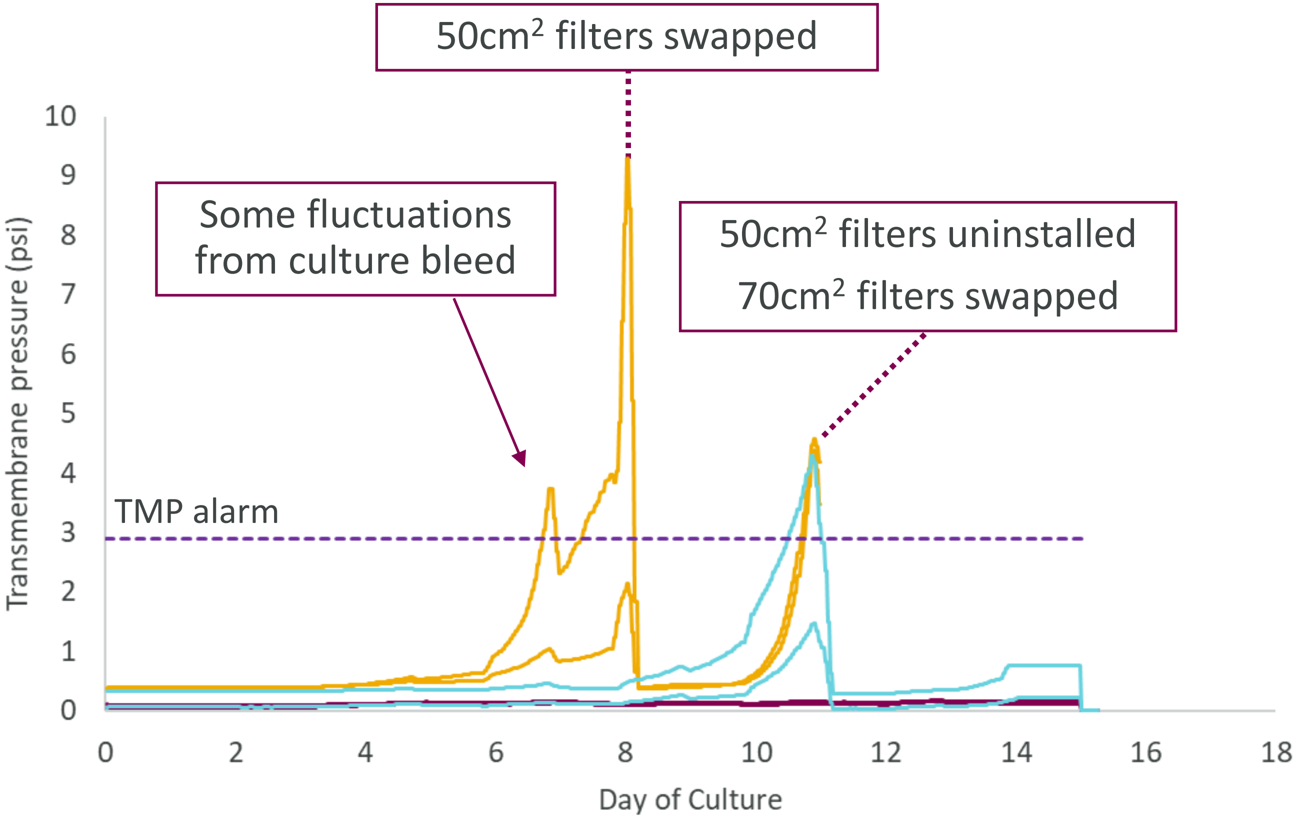




UMP filters lasted longer than in Experiment 1 and were successfully swapped

Bioreactors with 50cm^2 UMP filters ended early due to flux decay on second filter

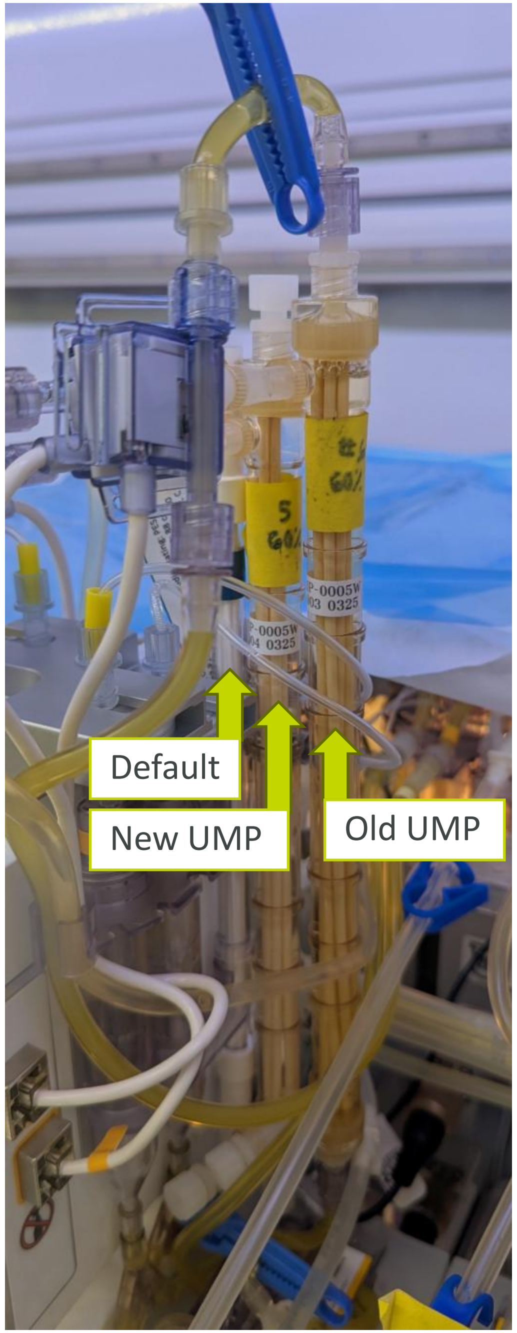




Logistics of a mid-run filter swap

Mid-run filter swap can be programmed in the Runtime software

1. Prior to filter swap, recirculation loop is emptied of culture*
2. New filter moved into position
3. All tubing clamped
4. Tubing disconnected at luer locks and connected to new filter:
   A. Top of filter
   B. Bottom of filter
   C. Permeate
5. Recirculation restarted – filter is automatically emptied into the bioreactor before recirculation resumes. Shell side of filter remains filled
6. New filter shell side is primed with media at user-specified flow rate



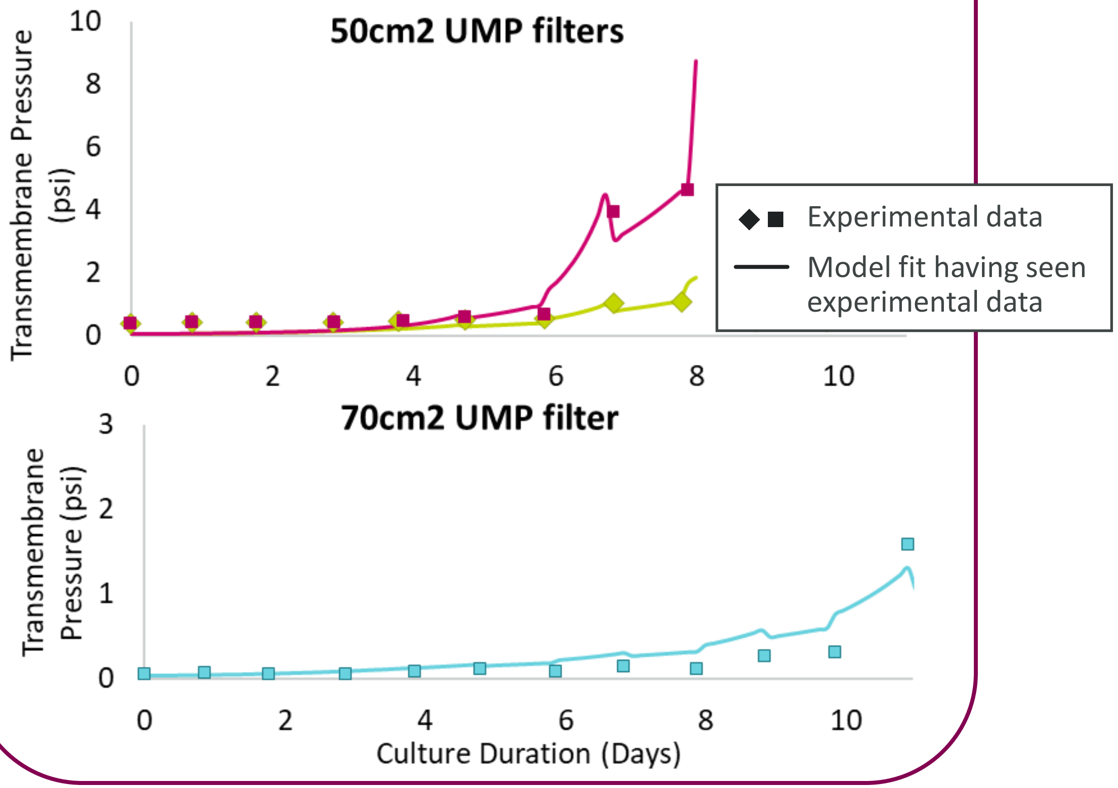
*For this experiment, the filter was swapped outside of the "Change filter assembly" menu and the old filter was not emptied of culture before swap

Can we predict filter performance across scales?

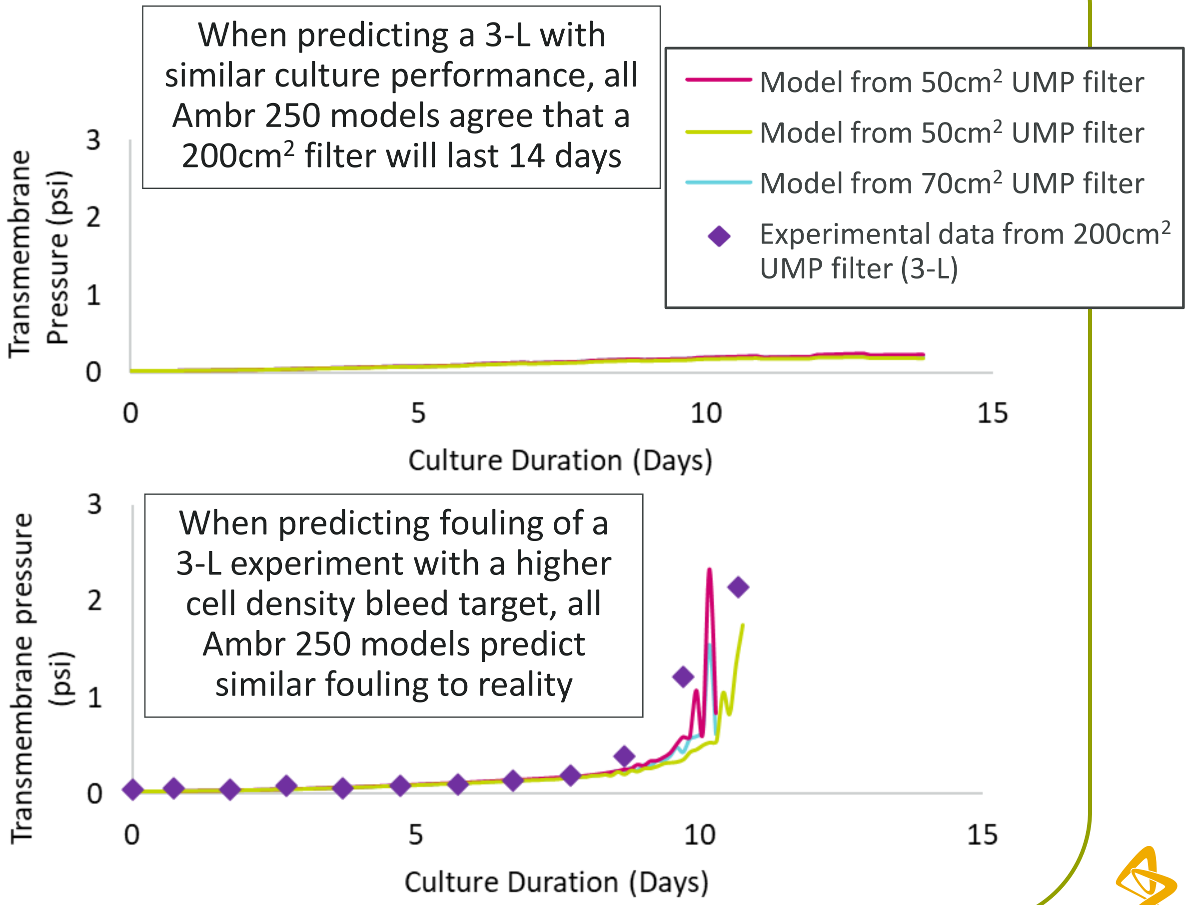
Model parameters are derived from Ambr 250 culture data and applied to 3-L scale

1. Experimental **data is ingested** by the model

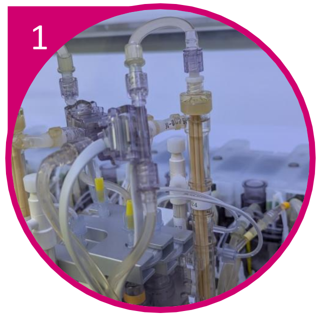
One set of coefficients is generated per bioreactor
These model fits have "seen" the experimental data



2. Can we **predict across scales** using models trained on Ambr 250 data?



Conclusions



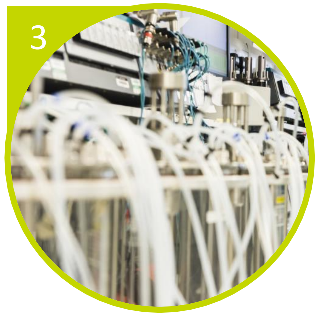
Filters were **successfully attached and swapped** with no resulting contaminations.

Filter reached acceptable height below liquid handler arm.

Filters can be attached and swapped in place, likely easier to do prior to bioreactor installation.

No impact on cell culture performance due to filter type or filter packing solution.

Flux decay can be identified by spiking transmembrane pressure with enough notice to swap filters.



Poor model prediction for Cell Line A **likely impacted by shear rate discrepancy or filter drying**: follow-up study required to confirm flux decay for Cell Line A is predictable.

Promising results from Cell Line B showed **filter flux decay could be predicted** across scales and with different culture performance.

Acknowledgements

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